Product Requirements Document (PRD)

1. Introduction

Context:
Fraud continues to rise in eCommerce and digital payments. While businesses have fraud detection engines, many lack a transparent, real-time interface that empowers analysts to monitor suspicious transactions and act quickly.

Vision:
Develop a web-based Fraud Monitoring Dashboard that provides analysts and managers with real-time visibility into suspicious activity, streamlined triage tools, and clear reporting.

Problem Statement:
Existing fraud detection tools are either too slow, too opaque, or too rigid, leading to missed fraud attempts and wasted analyst time. A tailored dashboard will close this gap.

Goals:

- Enable fraud analysts to detect and act on suspicious transactions faster.
- Reduce false positives and false negatives through improved visibility.
- Provide fraud managers with insights into fraud patterns and case resolution trends.

Link to Strategy:
Supports the broader organizational goal of reducing fraud losses, increasing trust in the platform, and improving operational efficiency for fraud teams.

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2. Objectives & Success Metrics

- **Detection Effectiveness:** Identify at least 80% of high-risk fraud attempts before fulfillment.
- **Efficiency:** Reduce average triage time per case from 12 minutes to under 5 minutes.
- **Accuracy:** Reduce false positives by 20% within 6 months.
- **Adoption:** 90% of fraud analysts using the dashboard daily within 3 months of launch.

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3. Scope

In-Scope Features:

- Transaction ingestion via API or batch upload.
- Real-time fraud scoring and flagging.
- Dashboard UI with search, filters, and alerts.

- Case triage workflow (assign, resolve, escalate).
- Role-based reporting for managers.

Out-of-Scope Features (future work):

- Direct payment processor integrations.
- Automated chargeback dispute management.
- Full machine learning model training (initial version will use rules + basic scoring).

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4. User Stories & Use Cases

User Stories:

- As a fraud analyst, I want to see flagged transactions in real time so I can act before fulfillment.
- As a fraud analyst, I want to filter cases by score, payment method, and region so I can prioritize effectively.
- As a fraud manager, I want trend reports so I can evaluate fraud patterns and team performance.
- As an engineer, I want clear API documentation so I can integrate transaction feeds.

Edge Cases & Constraints:

- Transactions with missing or malformed fields must not crash the system.
- System must support at least 10,000 transactions/day in MVP.
- Analysts may have varying screen sizes; UI should adapt responsively.

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5. Functional Requirements

- **Must Have:**
    - Real-time ingestion of transactions.
    - Fraud scoring engine with thresholds.
    - Dashboard with alerts, filters, and case detail views.
    - Case management workflow (assign/resolve).
- **Should Have:**
    - Export of case data to CSV.
    - Manager reporting dashboard.

- **Could Have:**
    - Notification system (email/Slack alerts).
    - Customizable scoring thresholds.

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6. Non-Functional Requirements

- **Performance:** Dashboard pages load in under 2 seconds.
- **Scalability:** Support up to 100k transactions/day within 12 months.
- **Accessibility:** Compliant with WCAG 2.1 AA.
- **Security:** Enforce role-based authentication and encrypted data storage.
- **Compliance:** Ensure GDPR compliance for handling user data.

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7. Dependencies & Risks

Dependencies:

- Cloud hosting (AWS/Azure/GCP).
- External fraud scoring library or rules engine.
- CI/CD pipeline for deployment.

Risks & Mitigation:

- **Risk:** Delayed data feeds reduce dashboard usefulness.
    - *Mitigation:* Implement buffer queues with retry logic.
- **Risk:** Analysts may resist adoption due to learning curve.
    - *Mitigation:* Provide onboarding materials and training.
- **Risk:** Scaling issues with data ingestion.
    - *Mitigation:* Start with throttled MVP, add load testing before scaling.

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8. Acceptance Criteria

- A transaction flagged by the fraud engine appears in the dashboard within 5 seconds of ingestion.
- Analysts can filter cases by at least 3 attributes (e.g., score, payment type, region).
- Managers can generate weekly fraud reports from the dashboard.

- Case status (open, resolved, escalated) is tracked and updated correctly.
- Authentication system prevents unauthorized access to case data.

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9. Distinction from Related Documents

- **PRD (this document):** Defines what the product must do and why.
- **PDR (Product Design Record):** Will specify UX, wireframes, and architecture choices.
- **Technical Specification:** Will define implementation details at the code and API level.